Machine Learning-Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence

Research paper · Parcl Co. Limited (Unified Mentor project)

Abstract

We segment 2,000 property buyers of a multi-tower residential/office development using K-Means clustering, validated with Ward hierarchical clustering. Client attributes (age, satisfaction, client type, purchase purpose, loan use, referral channel, domestic/international) are joined to behavioural purchase features derived from 7,305 sold units (units bought, total spend, average price per sq ft). Four segments emerge: **Younger Satisfied**, **Senior Established**, **Dissatisfied Mid-Life** and a small **Multi-Unit Portfolio** group. The most actionable findings are that the dissatisfied group holds 27% of revenue, and that 51 portfolio buyers (2.5% of clients) generate 4.9% of it at twice the average ticket. Cluster separation is modest (silhouette 0.12) because the underlying attributes are nearly uncorrelated; we report this limitation explicitly and add a transparent rule-based persona overlay (Global Investor, First-Time, Corporate, Luxury) that mirrors the brief's business language.

1. Background and objectives

Without segmentation a developer markets identically to home buyers, institutions, foreign investors and first-time purchasers, which wastes marketing spend and produces generic recommendations. The objectives are to (i) discover hidden buyer groups, (ii) describe their investment motives, geography and financing behaviour, and (iii) deliver an interactive dashboard so marketing teams can explore segments by country, region, purpose and client type.

2. Data

Table	Rows	Content
clients.csv	2,000	client_id, client_type, name, date_of_birth, gender, country, region, acquisition_purpose, satisfaction_score (1-
properties.csv	10,000	listing_id, tower, transaction_date, unit_category (Apartment/Office), floor_area_sqft, sale_price, listing_status

2.1 Cleaning

• No missing client attributes and no duplicate client_id or listing_id were found (missing cells: 0, duplicate listings: 0); the pipeline nonetheless strips whitespace, normalises label case and de-duplicates.

• **Mixed date formats:** date_of_birth mixes dd-mm-yyyy (dash) and m/d/yyyy (slash). Both were parsed explicitly (0 unparsed). Age is measured at the last transaction date.

• sale_price was stored as text ("$300,385.62") and converted to float.

• Only *Sold* listings with a client_ref were attributed to buyers; all 2,000 clients own at least 3 units (max 13). Unit numbers repeat within a tower (e.g. two listings of unit 4), which we treat as data-entry noise and did not use.

2.2 Feature engineering

Per-client aggregates from sold units: **units_purchased**, **total_spend**, avg_price, avg_area, **avg_ppsf** (price per sq ft), office_share. Derived client fields: age, age group and **is_international** (country != USA). These are the behavioural signals the brief calls for (“multiple units”, “large investments”), which the raw client table cannot provide.

3. Exploratory data analysis

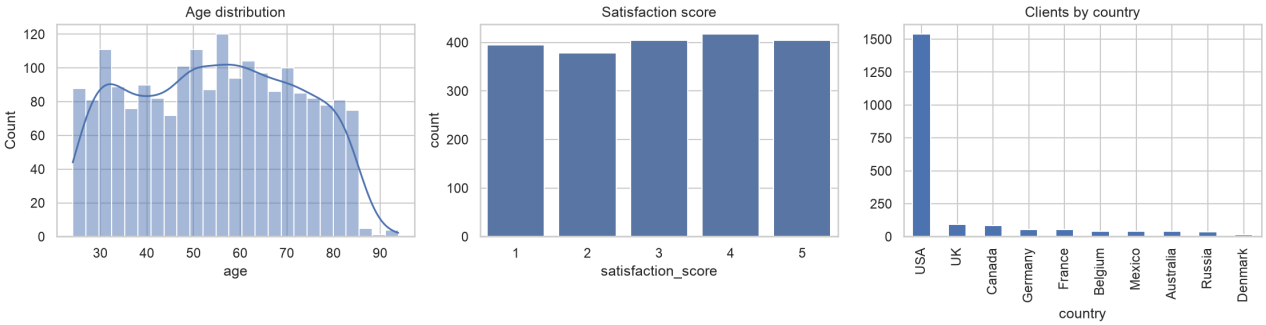


Figure 1. Univariate: ages span 24-94, satisfaction is nearly uniform, and 77% of clients are in the USA (California alone is 32%).



Figure 2. Bivariate: purpose, loan use and channel are almost identical for individuals and companies.

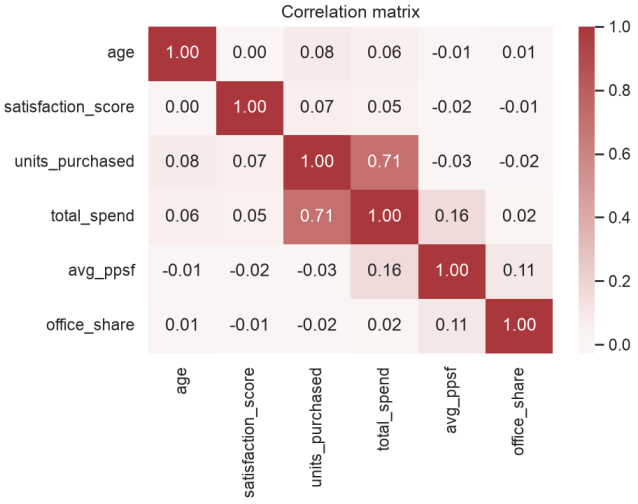


Figure 3. Multivariate: only units_purchased vs. total_spend (0.71) is strongly related; client attributes are near-independent.



Figure 4. Market context: listing volume by month, price distribution by unit category, and price vs. area.

Key observations. (1) 31% of buyers purchase for investment, 37% apply for a loan, 5.1% are companies. (2) Offices sell at a ~6% higher mean price ($362k vs $341k for apartments). (3) Correlations between any two client attributes are below 0.12 in absolute value: investors are not systematically older, wealthier or more loan-dependent than home buyers. This is the single most important fact for interpreting the clustering.

4. Methodology

Encoding and scaling. One-hot encoding for client_type, acquisition_purpose, loan_applied, referral_channel and is_international (country and region are one-hot-heavy with 10 and 57 levels, so they are collapsed to domestic/international for clustering and used for post-hoc geographic profiling). StandardScaler for age, satisfaction_score, units_purchased, total_spend and avg_ppsf, all wrapped in a scikit-learn ColumnTransformer saved as models/pipeline.pkl.

Models. K-Means (n_init=50, seed 42) as the primary, interpretable model; agglomerative (Ward) clustering as an independent check and to reveal nested structure.

Choosing k.

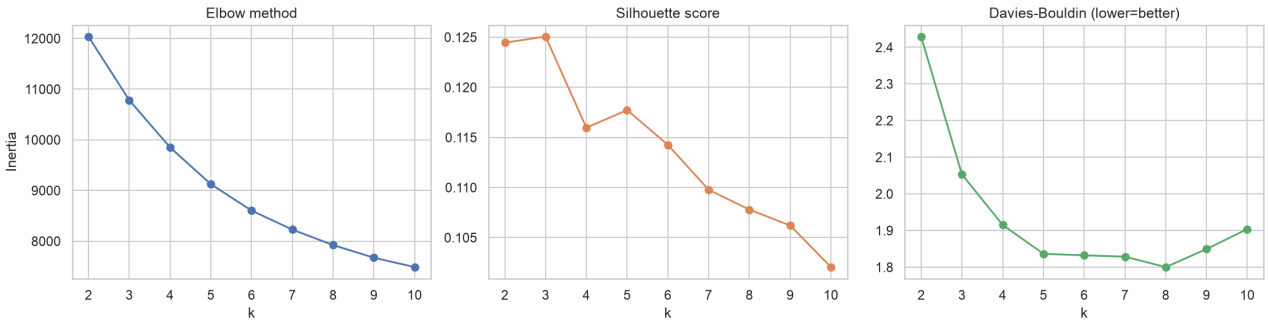


Figure 5. Elbow shows no sharp bend; silhouette is flat (0.10-0.125) for k=2-10; Davies-Bouldin improves steadily until k=6-8.

k	Inertia	Silhouette	Davies-Bouldin
2	12,038	0.124	2.43
3	10,782	0.125	2.05
4	9,852	0.116	1.92
5	9,125	0.118	1.84
6	8,605	0.114	1.83
7	8,228	0.110	1.83
8	7,924	0.108	1.80
9	7,673	0.106	1.85
10	7,488	0.102	1.90

No k is statistically dominant. We chose **k = 4** because (a) silhouette at k=4 (0.116) is within 0.01 of the maximum, (b) the brief recommends four segments, and (c) four groups remain distinct and actionable, whereas larger k mostly splits groups on noise. This is a judgement call, not a statistical optimum.

5. Results

5.1 Segment profiles

Segment	Clients	Age	Satisf.	Invest.	Loan	Corp.	Intl.	Units	Avg spend	% revenue
Younger Satisfied Buyers	701	42.6	4.12	33%	40%	8%	26%	3.5	$1.24M	35%
Senior Established Buyers	689	72.7	3.19	27%	34%	1%	21%	3.6	$1.23M	34%

Segment	Clients	Age	Satisf.	Invest.	Loan	Corp.	Intl.	Units	Avg spend	% revenue
Dissatisfied Mid-Life Buyers	559	46.8	1.42	33%	37%	7%	23%	3.5	$1.21M	27%
Multi-Unit Portfolio Buyers	51	62.4	3.45	33%	35%	4%	14%	7.3	$2.42M	5%

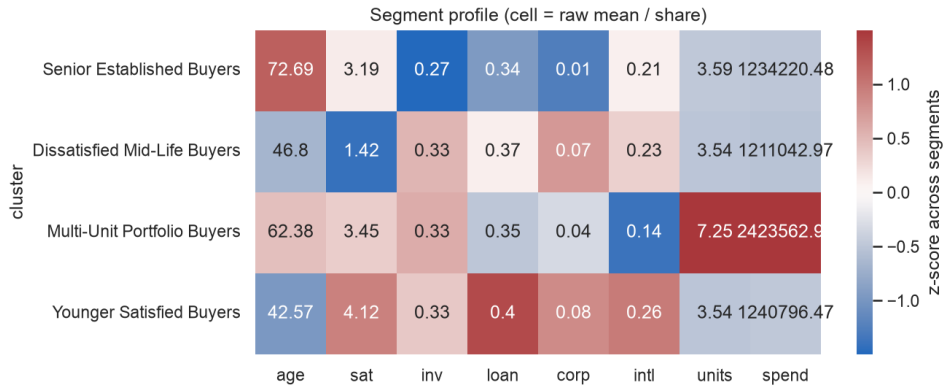


Figure 6. Segment profile heat-map (colour = z-score across segments, text = raw value).

• **Multi-Unit Portfolio Buyers** (51 clients): 7.3 units and $2.4M each, ~2x the average; the genuine “large investor” group, found by purchase behaviour, not by the self-declared investment flag.

• **Dissatisfied Mid-Life Buyers** (559): mean satisfaction 1.42/5, age ~47. Purchasing is typical, so this is a retention and reputation risk worth 27% of revenue.

• **Senior Established Buyers**: oldest (mean age 73), neutral-to-positive satisfaction, almost no companies (0.9%), lowest investment share (27%).

• **Younger Satisfied Buyers**: youngest (~43), satisfaction 4.1, highest loan use (40%) and international share (26%).

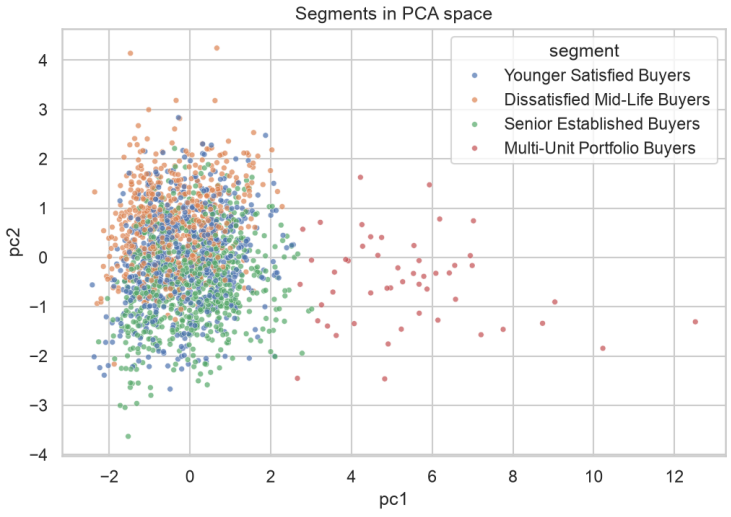


Figure 7. Segments in PCA space: boundaries overlap, consistent with the low silhouette.

5.2 Hierarchical validation

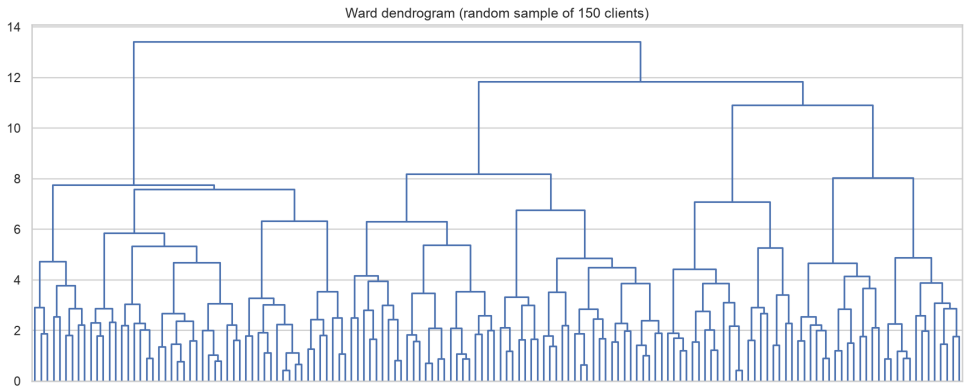


Figure 8. Ward dendrogram on a 150-client sample.

Ward clustering with k=4 gives silhouette 0.065 versus 0.116 for K-Means, and an adjusted Rand index of 0.27 between the two solutions. Agreement is only partial: the dendrogram shows no deep split, so the cluster boundaries are not natural gaps. K-Means is retained for its stability and interpretability; the segments should be treated as useful, soft groupings.

5.3 Persona overlay aligned to the brief

The brief names Global Investors, First-Time Buyers, Corporate Buyers and Luxury Investors. Because the data do not produce these as separate clusters, we tag them with explicit, auditable rules (Corporate: client_type=Company; Luxury: top-decile total spend and satisfaction ≥ 4; Global: investment purpose and non-US; First-Time: home purpose, loan, age < 40; otherwise Mainstream), applied in that priority order.

Persona	Clients	Mean age	Satisf.	Units	Avg spend
Mainstream Buyer	1564	56.7	2.96	3.6	$1.23M
Global Investor	128	54.7	2.95	3.5	$1.22M
First-Time Buyer	121	31.7	2.94	3.5	$1.20M
Corporate Buyer	103	46.6	3.07	3.7	$1.26M
Luxury Investor	84	59.3	4.49	5.2	$1.98M

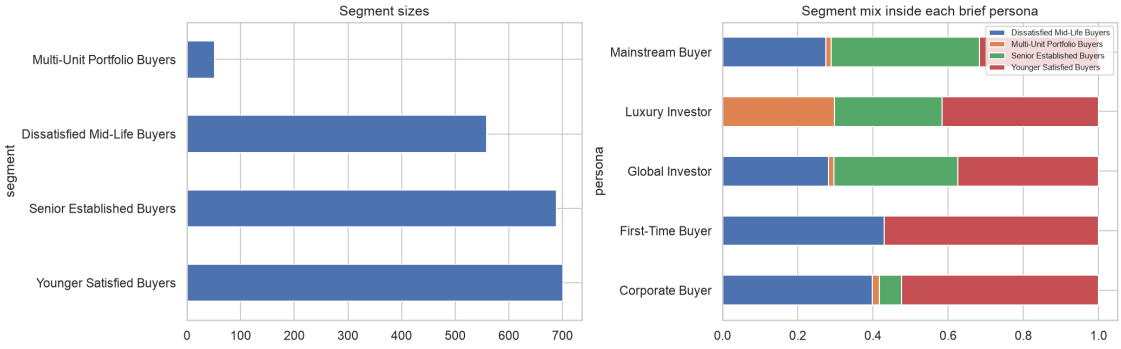


Figure 9. Segment sizes and the cluster mix inside each persona.

6. Streamlit dashboard

The app (app.py) provides four views: **Segmentation overview** (cluster distribution, PCA map, persona counts); **Investor behaviour** (purpose, loan, spend, units and channel by segment); **Geographic analysis** (country choropleth, dominant segment by US state, region-by-segment heat-map); and **Segment insights** (per-segment KPIs, descriptive statistics, recommended action, CSV export). Sidebar filters cover country, region, acquisition purpose and client type, and every chart responds to them.

7. Recommendations

• **Service recovery for the Dissatisfied segment.** Contact the 559 clients rated ≤ 2, find the root cause (build quality, handover, communication) and track satisfaction quarterly; $677M of purchase value sits here and unhappy buyers depress referrals (only 10% of clients arrive by referral).

• **Account management for Portfolio Buyers.** A dedicated relationship manager, bulk or bundle pricing and pre-launch access for 51 clients who spend $2.4M each. Lookalike targeting should use units purchased, not the self-reported investment flag.

• **Financing-led offers for younger buyers,** who use loans most; partner with lenders and highlight monthly-payment framing.

• **Do not segment marketing on purpose or country alone:** these carry almost no predictive signal in this data (Section 3). Collect richer fields (income, source of funds, prior ownership, property use, budget) to make future clustering sharper.

• **Geographic focus:** California is a third of clients; test international campaigns on the UK/Germany/France/Canada tail, where average spend is comparable to the US.

8. Limitations

• Weak cluster structure (silhouette 0.12; low K-Means/Ward agreement). Segments are descriptive, not sharply separated.

• Client attributes appear statistically independent, which suggests synthetic or heavily anonymised data; findings may not transfer to live data.

• No income, wealth or first-time-buyer field exists, so personas rely on proxies (age, loan, spend).

• Satisfaction is a single 1-5 score with no timestamp, so causal claims are not supported.

9. Conclusion

Behavioural purchase features reveal a small, high-value portfolio-buyer group and a large satisfaction-defined risk group that demographic fields alone would miss. The work delivers a reproducible pipeline, a saved model and a filterable dashboard, and is candid that stronger segmentation requires richer buyer data.